Klasifikasi Dokumen Multimodal: Penggabungan Fitur Teks Hasil OCR dan Citra Dokumen

Disusun Oleh:

Yulian Anggreaif Rudamaga : 256150100011001
Zalzabila Rani : 256150100011008
Ramadhani Al-Qadri : 256150100011010

Multimodal deep networks for text and image-based document classification

Nicolas Audebert Catherine Herold Kuider Slimani Cédric Vidal

Quicksign, 38 rue du Sentier, 75002 Paris

{nicolas.audebert,catherine.herold,kuider.slimani,cedric.vidal}@quicksign.com

Résumé

La classification automatique de documents numérisés est importante pour la dématérialisation de documents historiques comme de procédures administratives. De premières approches ont été suggérées en appliquant des réseaux convolutifs aux images de documents en exploitant leur aspect visuel. Toutefois, la précision des classes demandée dans un contexte réel dépend souvent de l'information réellement contenue dans le texte, et pas seulement dans l'image. Nous introduisons un réseau de neurones multimodal capable d'apprendre à partir d'un plongement lexical du texte extrait par reconnaissance de caractères et des caractéristiques visuelles de l'image. Nous démontrons la pertinence de cette approche sur Tobacco3482 et RVL-CDIP, augmentés de notre jeu de données textuel QS-OCR[1], sur lesquels nous améliorons les performances d'un modèle image de 3% grâce à l'information sémantique textuelle.

Mots-clés

Classification de documents, apprentissage multimodal, fusion de données.

Abstract

Classification of document images is a critical step for archival of old manuscripts, online subscription and administrative procedures. Computer vision and deep learning have been suggested as a first solution to classify documents based on their visual appearance. However, achieving the fine-grained classification that is required in real-world setting cannot be achieved by visual analysis alone. Often, the relevant information is in the actual text content of the document. We design a multimodal neural network that is able to learn from word embeddings, computed on text extracted by OCR, and from the image. We show that this approach boosts pure image accuracy by 3% on Tobacco3482 and RVL-CDIP augmented by our new QS-OCR text dataset[1], even without clean text information.

Keywords

Document classification, multimodal learning, data fusion.

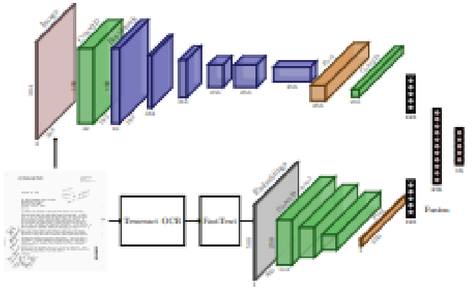


Figure 1: Multimodal classifier for hybrid text/image classification. Training is performed end-to-end on both textual and visual features.

The ubiquity of computers and smartphones has incentivized governments and companies alike to digitize most of their processes. Onboarding new clients, paying taxes and proving one's identity is more and more done through a computer, as the rise of online banking has shown in the last few years. Industrial and public archives are also ongoing serious efforts to digitize their content in an effort for preservation, e.g. for old manuscripts, maps and documents with a historical value. This means that previously physical records, such as forms and identity documents, are now digitized and transferred electronically. In some cases, those records are produced and consumed by fully automated systems that rely on machine-readable formats, such as XML or PDF with text layers. However, most of these digital copies are generated by end-users using whatever mean they have access to, i.e. scanners and cameras, especially from smartphones. For this reason, human operators have remained needed to proofread the documents, extract selected fields, check the records' consistency and ensure that the appropriate files have been submitted. Automation through expert systems and machine learning can help accelerate this process to assist and alleviate the burden of this fastidious work for human workers.

A common task involved in data filing processes is document

Latar Belakang & Ide Utama

1 Dokumen = Bimodal

Satu halaman mengandung informasi visual (Layout, tabel, tanda tangan) dan tekstual (kata kunci)

2 Masalah Unimodal

Citra bisa bingung pada layout yang mirip.
Teks gagal ketika hasil OCR buruk

3 Solusi Multimodal

Menggabungkan fitur citra + teks OCR supaya informasi yang saling melengkapi dapat dimanfaatkan

- Penelitian mereplikasi arsitektur dua cabang. Audebert et al. (2019).
- Eksperimen menggunakan dataset Tobacco 3482 dengan dua modalitas.
- Dua strategi fusion dibandingkan untuk melihat metoda yang paling efektif.

Problem Statement

1. Seberapa besar akurasi naik ketika kedua modalitas digabung, dibanding memakai satu sumber terbaik saja?

2. Dari dua cara penggabungan yang diuji, mana yang benar-benar bekerja?

3. Apakah model gabungan sungguh memakai kedua sumber, atau sebenarnya hanya mengandalkan satu saja?

Modalitas

Aspek	Citra Dokumen	Teks Hasil OCR
Representasi masukan	Grayscale satu halaman penuh, 384 x 384, tanpa padding (rasio aspek di warp)	Sekuens kata, zero-padded sampai 500 kata
Ekstraksi	-	Tesseract 4.0, mesin LSTM (--oem 1 –psm 3 -1eng)
Encoder Kata	-	FastText 300dimensi, pralatih Common Crawl
Arsitektur	MobileNetV2, bobot awal ImageNet, kanal grayscale diduplikasi tiga kali	CNN 1D bergaya Kim, 4 lapis, filter multi-lebar, max-pooling-through-time
Dimensi fitur keluaran	128 (dari 1280 hasil global average pooling dipetakan lapisan fully connected	128

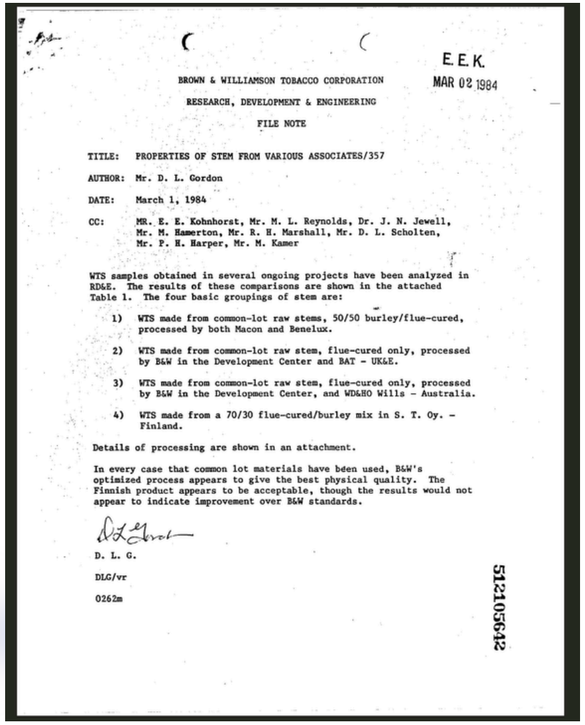
Dataset

Tobacco3482-jpg

3.482 citra - 10 kelas

Sumber: Kaggle (patrickaudriaz/tobacco3482jpg)

Status Lisensi: Tidak ada lisensi baku yanng dinyatakan



E.E.K.

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MAR 02 1984

RESEARCH, DEVELOPMENT & ENGINEERING

FILE NOTE

TITLE: PROPERTIES OF STEM FROM VARIOUS ASSOCIATES/357

AUTHOR: Mr. D. L. Gordon

DATE: March 1, 1984

CC: MR. E. E. Kohnhorst, Mr. M. L. Reynolds, Dr. J. N. Jewell, Mr. M. Hamerton, Mr. R. H. Marshall, Mr. D. L. Scholten, Mr. P. H. Harper, Mr. M. Kamer

WTS samples obtained in several ongoing projects have been analyzed in RD&E. The results of these comparisons are shown in the attached Table 1. The four basic groupings of stem are:

1) WTS made from common-lot raw stems, 50/50 burley/flue-cured, processed by both Macon and Benelux.

2) WTS made from common-lot raw stem, flue-cured only, processed by B&W in the Development Center and BAT - UK&E.

3) WTS made from common-lot raw stem, flue-cured only, processed by B&W in the Development Center, and WD&HO Wills - Australia.

4) WTS made from a 70/30 flue-cured/burley mix in S. T. Oy. - Finland.

Details of processing are shown in an attachment.

In every case that common lot materials have been used, B&W's optimized process appears to give the best physical quality. The Finnish product appears to be acceptable, though the results would not appear to indicate improvement over B&W standards.

D. L. G.

DLG/vr

0262m

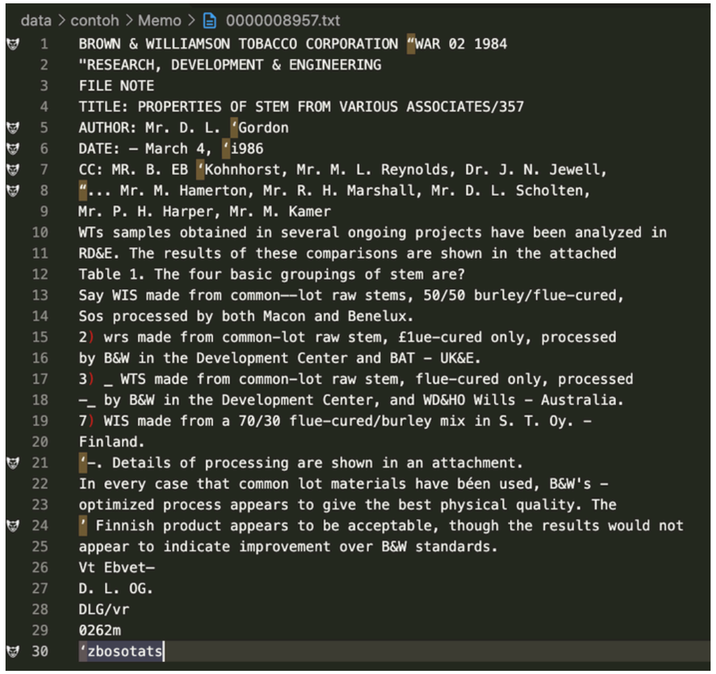
512105642

QS-OCR-Small

3.482 file .txt

Sumber: GitHub QuickSign/ocrized-text-dataset, rilis v1.0

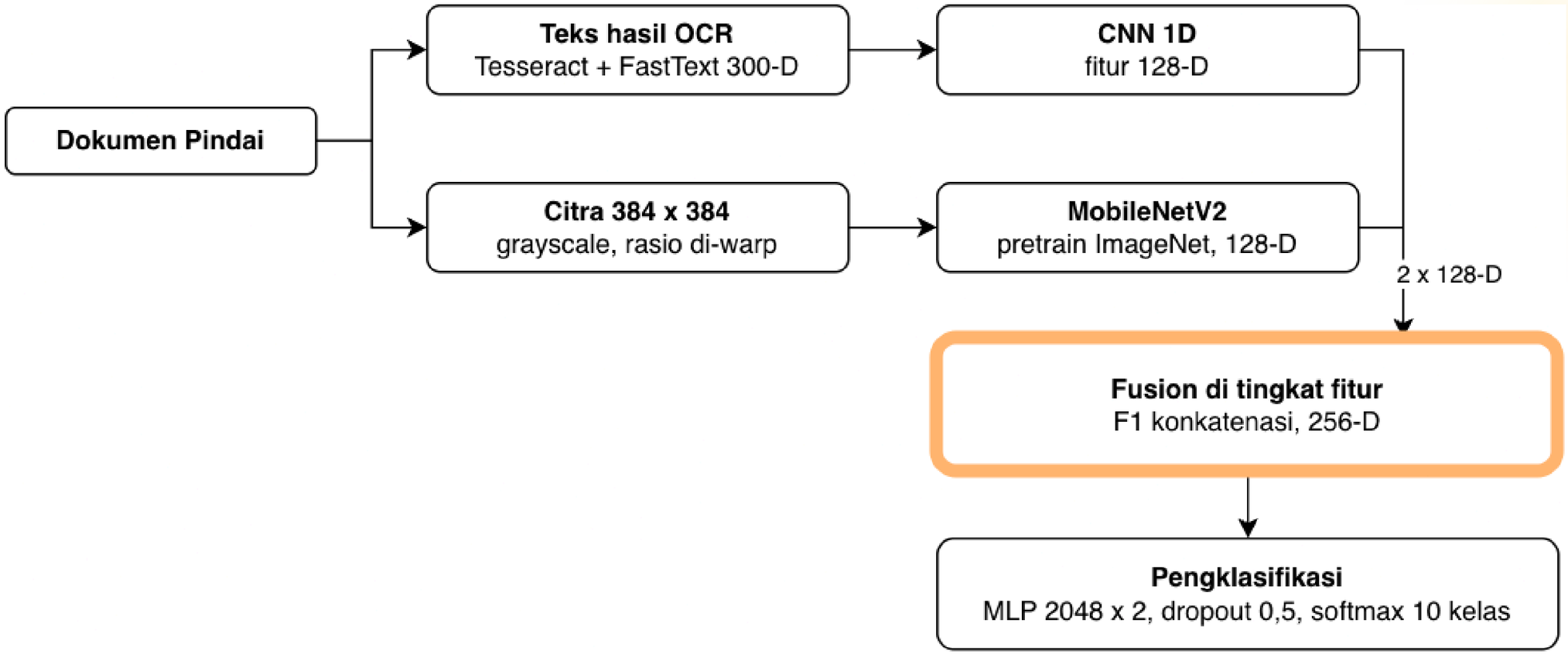
Lisensi Repositori: Apache-2.0; teks turunan Tobacco3482



data > contoh > Memo > 0000008957.txt

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1  BROWN & WILLIAMSON TOBACCO CORPORATION “WAR 02 1984
2  "RESEARCH, DEVELOPMENT & ENGINEERING
3  FILE NOTE
4  TITLE: PROPERTIES OF STEM FROM VARIOUS ASSOCIATES/357
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6  DATE: – March 4, ‘i986
7  CC: MR. B. EB ‘Kohnhorst, Mr. M. L. Reynolds, Dr. J. N. Jewell,
8  “... Mr. M. Hamerton, Mr. R. H. Marshall, Mr. D. L. Scholten,
9  Mr. P. H. Harper, Mr. M. Kamer
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26 Vt Ebvet–
27 D. L. OG.
28 DLG/vr
29 0262m
30 ‘zbosotats
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Metode Fusion: Late Fusion



Metrik Evaluasi

1 Overal accuracy (OA)

Metrik utama paper; perlu agar perbandingan langsung dengan angka yang dilaporkan menjadi mungkin.

2 Macro F1

Kelas timpang (~5:1 terbesar berbanding terkecil). OA sendirian didominasi kelas besar, jadi macro F1 dilaporkan berdampingan, tidak menggantikan.

3 F1 per kelas + confusion matrix

Keuntungan fusion diperkirakan tidak merata; kelas mana yang membaik menunjukkan modalitas mana menyumbang apa.

4 Oracle (batas atas)

Benar bila cabang teks atau citra benar. Memungkinkan keuntungan fusion dinyatakan sebagai persentase potensi yang dipanen, bukan selisih mentah.

5 Missing-modality test

Penurunan akurasi saat satu modalitas dinolkan pada inferensi. Satu-satunya metrik yang dapat membedakan fusion sejati dari dominasi satu cabang.

6 Ablasi degradasi bertingkat

Rotasi, blur, derau, dan kompresi JPEG pada citra; penghapusan kata dan penyuntikan salah eja pada teks. Menguji klaim ketahanan yang tidak diuji paper.